Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 01)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows three investigations carried out by a class.

Table 1

Investigation	Data collected
A	Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed)
B	Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread
C	Soil moisture and the percentage cover of a moss, measured at 14 points along a transect

0 1 . 1 Name the most suitable statistical test for each investigation in Table 1.

[3 marks]

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0 1 . 2 Explain what it means if a test gives a probability of $p = 0.02$.

[2 marks]

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0 1 . 3 Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

[3 marks]

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0 1 . 4 Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

[3 marks]

0 1 . 5 The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

[4 marks]

15

0 2 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☒

D To measure the concentration of a pollutant at the point of release ☐

0 2 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 2 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 3 Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg^{-1}
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

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0 3 . 1 Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

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0 3 . 2 Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer ☐ tonnes

0 3 . 3 Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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0 3 . 4 Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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0 3 . 5 Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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0 4 A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 4 . 1 Calculate the percentage water content of the fresh soil.

[2 marks]

Answer ..... %

0 4 . 2 Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer ..... %

0 4 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 4 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 4 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 5 A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million m^3 of water. The density of water is 1000 kg m^{-3} and g = 9.8 N kg^{-1}.

0 5 . 1 Describe how pumped-storage HEP is used for peak shaving.

[3 marks]

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0 5 . 2 Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

[3 marks]

Answer ☐ J

0 5 . 3 The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

[2 marks]

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0 5 . 4 Give **two** locational requirements of a pumped-storage station.

[2 marks]

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0 6 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 6 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 6 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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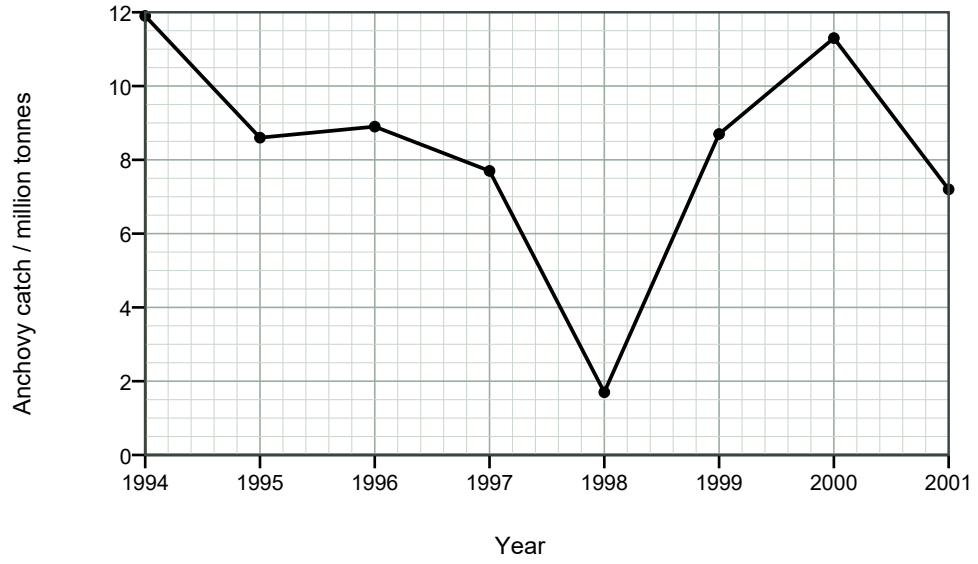
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0 7 Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



0 7 . 1 Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 7 . 2 Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

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0 7 . 3 Describe **two** other effects of El Niño events.

[2 marks]

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0 7 . 4 Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

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0 8 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 8 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 8 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 8 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 9 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 9 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

Answer ..... %

0 9 . 2 Explain how each of the following reduces heat loss.

[3 marks]

Cavity wall insulation .....

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Low-emissivity double glazing .....

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A heat exchanger on the ventilation system .....

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0 9 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

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0 9 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

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0 9 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

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1 0 . 1 Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

[9 marks]

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Extra space

1 0 . 2 State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS